

Assignment-3

Task:1

Problem-01:

-

A bit stream 1101011011 is transmitted using the standard CRC method. The generator polynomial is x^4+x+1 . What is the actual bit string transmitted?

```

task01.cpp > xorDivision(string, string)
1  #include <iostream>
2  #include <string>
3  using namespace std;
4
5  // Function to perform XOR division and find remainder
6  string xorDivision(string data, string generator) {
7      int n = generator.length();
8      string temp = data.substr(0, n);
9
10     while (n <= data.length()) {
11         if (temp[0] == '1') {
12             for (int i = 1; i < generator.length(); i++)
13                 temp[i] = (temp[i] == generator[i]) ? '0' : '1';
14         }
15         temp.erase(0, 1);
16         if (n < data.length())
17             temp += data[n];
18         n++;
19     }
20     return temp;
21 }
22
23 int main() {
24     string data = "1101011011";
25     string generator = "10011"; //  $x^4 + x + 1$ 
26
27     // Append zeros equal to degree of generator
28     int m = generator.length() - 1;
29     string appendedData = data + string(m, '0');

```

```

30
31 // Get remainder using XOR division
32 string remainder = xorDivision(appendData, generator);
33
34 // Transmitted message = data + remainder
35 string transmitted = data + remainder;
36
37 cout << "Original Data: " << data << endl;
38 cout << "Generator: " << generator << endl;
39 cout << "Remainder: " << remainder << endl;
40 cout << "Transmitted Data: " << transmitted << endl;
41
42 return 0;
43 }

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```

PS C:\Users\LENOVO\Desktop\OSCN> .\task01.exe
Original Data: 1101011011
Generator: 10011
Generator: 10011
Remainder: 1110
Transmitted Data: 1101011011110
PS C:\Users\LENOVO\Desktop\OSCN> 

```

Task:2

Problem-02:

A bit stream 10011101 is transmitted using the standard CRC method. The generator polynomial is x^3+1 .

1. What is the actual bit string transmitted?
2. Suppose the third bit from the left is inverted during transmission. How will receiver detect this error?

```
task2.cpp > mod2div(string, string)
1  #include <iostream>
2  #include <string>
3  using namespace std;
4
5  string xor1(string a, string b) {
6      string result = "";
7      for (int i = 1; i < b.length(); i++) {
8          result += (a[i] == b[i]) ? '0' : '1';
9      }
10     return result;
11 }
12
13 string mod2div(string dividend, string divisor) {
14     int pick = divisor.length();
15     string tmp = dividend.substr(0, pick);
16     int n = dividend.length();
17
18     while (pick < n) {
19         if (tmp[0] == '1')
20             tmp = xor1(divisor, tmp) + dividend[pick];
21         else
22             tmp = xor1(string(pick, '0'), tmp) + dividend[pick];
23         pick++;
24     }
25
26     if (tmp[0] == '1')
27         tmp = xor1(divisor, tmp);
28     else
29         tmp = xor1(string(pick, '0'), tmp);
30
31     return tmp;
32 }
33
34 int main() {
35     string data = "10011101";
36     string divisor = "1001"; // x^3 + 1
37 }
```

```

38 // Step 1: Append zeros
39 string appended_data = data + string(divisor.length() - 1, '0');
40
41 // Step 2: Get remainder
42 string remainder = mod2div(appended_data, divisor);
43
44 // Step 3: Form transmitted code
45 string transmitted = data + remainder;
46
47 cout << "Original Data:      " << data << endl;
48 cout << "Generator:          " << divisor << endl;
49 cout << "Appended Data:       " << appended_data << endl;
50 cout << "CRC Remainder:      " << remainder << endl;
51 cout << "Transmitted Codeword: " << transmitted << endl;
52
53 // Step 4: Simulate error (invert 3rd bit)
54 string received = transmitted;
55 received[2] = (received[2] == '0') ? '1' : '0'; // flip 3rd bit
56 cout << "\nReceived (with error): " << received << endl;
57
58 // Step 5: Check remainder at receiver
59 string remainder_receiver = mod2div(received, divisor);
60 cout << "Remainder at Receiver: " << remainder_receiver << endl;
61
62 if (remainder_receiver.find('1') != string::npos)
63 |   cout << "Error detected during transmission!" << endl;
64 else
65 |   cout << "No error detected." << endl;
66
67 return 0;
68 }

```

```
-teytrqwg.1dk' '--dbgExe=C:\MinGW\bin\gdb.exe' '--interpreter=mi'
```

```
PS C:\Users\LENOVO\Desktop\OSCN> .\task2.exe
```

```
Original Data:      10011101
```

```
PS C:\Users\LENOVO\Desktop\OSCN> .\task2.exe
```

```
Original Data:      10011101
```

```
Original Data:      10011101
```

```
Generator:          1001
```

```
Appended Data:      10011101000
```

```
CRC Remainder:      100
```

```
Transmitted Codeword: 10011101100
```

```
Received (with error): 10111101100
```

```
Remainder at Receiver: 100
```

```
CRC Remainder:      100
```

```
Transmitted Codeword: 10011101100
```

```
Received (with error): 10111101100
```

```
Remainder at Receiver: 100
```

```
Remainder at Receiver: 100
```

```
Error detected during transmission!
```

```
PS C:\Users\LENOVO\Desktop\OSCN> 
```